Technische Universität Dresden

Data Visualization *Course Work*

1 – Dear Data (Solutions)

Faculty of Computer Science – Institute of Software and Multimedia Technology



Computergraphik und Visualisierung



INTERACTIVE MEDIA LAB DRESDEN

2026


Tree of emotions
Number of photos of different focus (3 week)
〈DOORS I WALK THROUGH〉
[A week of door passages]
MON TUE WED THU FRI SAT SUN
Assignment of Data Visualization
Where does my € 70 Go ?
Feeling I had after listening to different song
Haribo Gummy Bear Color Count (Sample 5 packs)
Do Notifications
outlier
general trend
notifications
Screen time / hours
Feelings
happy
relaxed
stressed
neutral
Lost
Size = Spend time
Four drinks, seven days
Daily volume by beverage, in millilitres
Plain water
Apple spritzer
Milk
Other (cola)
Mon Tue Wed Thu Fri Sat Sun
Illustrative week based on typical portions. Empty slots indicate zero intake.
Combinations

Course Work – Assignment 1
Dear Data Topics

Drinks drank
Feelings felt (after listening to music)
Games gamed (incl. 2 different FFVII)
Haribo Gummy Bear Color Count
Do notifications lead to more screen time?
Doors I walk through
Languages spoken by people met
Number of photos of different focus
Top 5 Germany States by Population
Tree of emotions
Daily activities vs Feelings
…
7/8 perfect score -- great job, thank you!



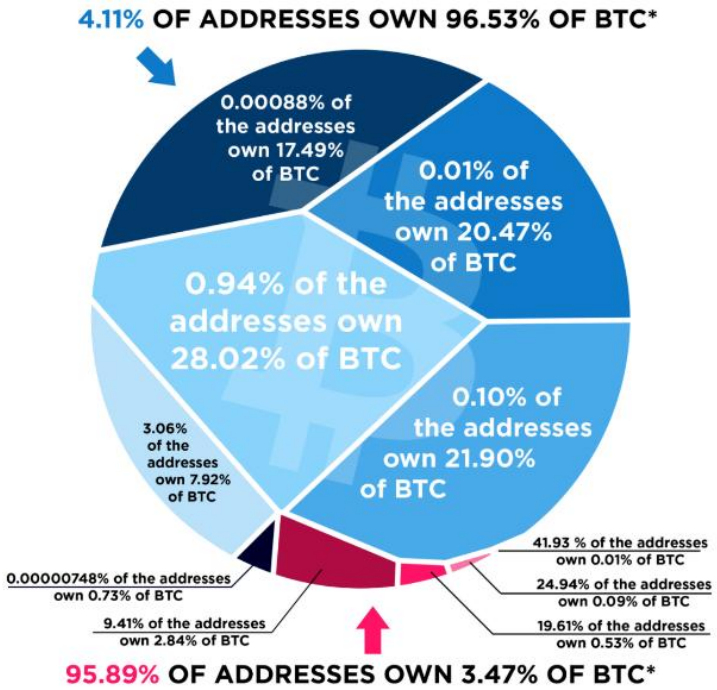
Course Work – Assignment 1
2.1 Lecture Recap: VisLies

- What's the main problem for each of the visualizations?

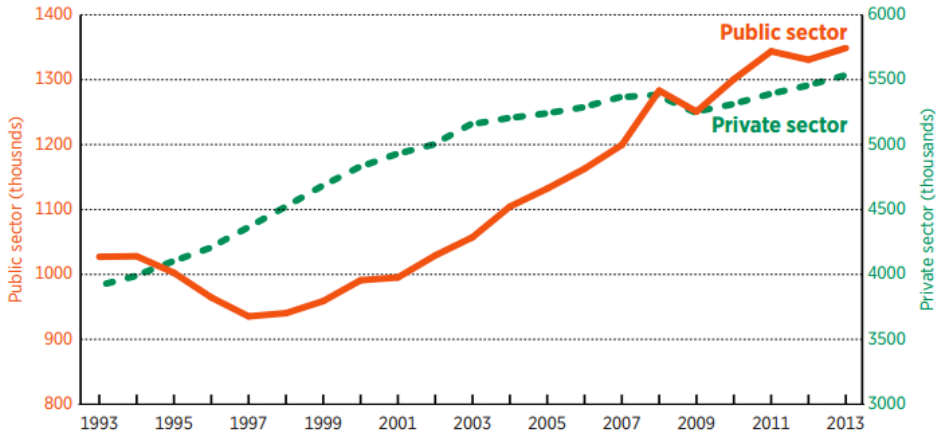
1. non-uniform polygon area used for quantitative + too much text (two values)
2. two uneven y axes scales, no obvious mapping

- How could you change the visualization to eliminate these problems?

1. less text, pie chart + legend, or even better, barchart
2. only one y axis - fair comparison



Source: https://howmuch.net/articles/bitcoin-wealth-distribution

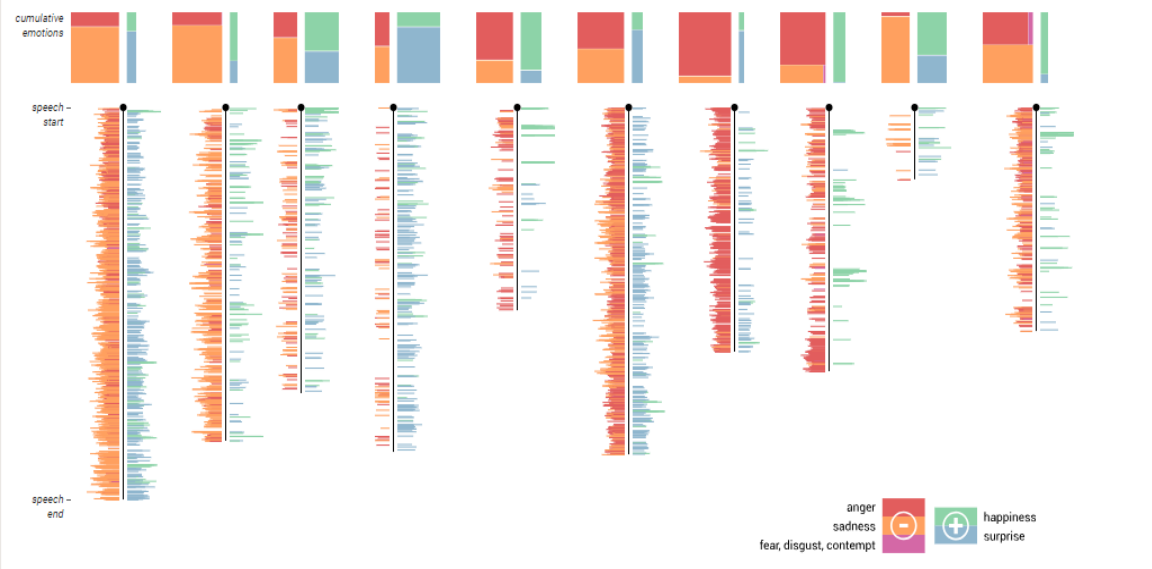


Source: https://www.fraserinstitute.org/sites/default/files/ontario-no-longer-a-place-toprosper.pdf

Course Work – Assignment 1
2.2 Lecture Recap: Perception and Color

Gestalt Principles:

- Which gestalt principles have been used in the visualization shown below?



similarity, proximity, closure (a bit symmetry)

Color Scales:

- Name & explain 3 problems with rainbow scales
- Give examples of better scales



cyclic, not ordered, poor contrast, unnatural in several regions

https://www.sciencedirect.com/science/article/pii/S0924271622002659

better ones: depends on data.

In general, high lightness contrasts = good color scale.
e.g. Grey scale, Parula, Viridis…

For more than 2 colors in a scale, maybe consider another visual variable :)